Year 13 Maths - Conditional probability

Similar questions - Tree diagrams and Venn diagrams

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Ravi and Sam play a match. The probability that Ravi wins any game is 0.6 and the probability that Sam wins is 0.4, independently of other games. The first player to win two games wins the match.

(a) Find the probability that the match lasts exactly 3 games. **[2]**

(b) Draw a tree diagram showing all possible outcomes of the match. **[3]**

(c) Find the probability that Sam wins the match. **[3]**

(d) Given that Sam wins the match, find the probability that it lasts exactly 3 games. **[3]**

2 There are 30 students in a class.

- 4 students study both Physics and Chemistry.
- 8 students study neither Physics nor Chemistry.
- The number who study Physics but not Chemistry is twice the number who study Chemistry but not Physics.

(a) Show this information in a Venn diagram, and find the probability that a randomly chosen student studies Chemistry. **[4]**

(b) Two different students are chosen at random. Given that exactly one of them studies Chemistry, find the probability that exactly one of them studies Physics. **[6]**

Question 1

(a) The match lasts 3 games when it is 1–1 after two games: $2 \times 0.6 \times 0.4 = 0.48$

(b) Branches: R then R (0.36, Ravi wins); R then S then R (0.144, Ravi) or S (0.096, Sam); S then R then R (0.144, Ravi) or S (0.096, Sam); S then S (0.16, Sam wins)

(c) P(Sam wins) $= 0.4^2 + 2(0.6)(0.4)^2 = 0.16 + 0.192 = 0.352$

(d) P(3 games and Sam wins) $= 0.192$. $\dfrac{0.192}{0.352} = \dfrac{6}{11} = 0.545$

Question 2

(a) Let C only $= x$, P only $= 2x$: $4 + 3x + 8 = 30$, $x = 6$. Venn: both 4, Chemistry only 6, Physics only 12, neither 8. P(Chemistry) $= \dfrac{10}{30} = \dfrac{1}{3}$

(b) Exactly one studies Chemistry: $10 \times 20 = 200$ pairs (one of the 10 Chemistry students, one of the 20 others). Of these, exactly one studies Physics when the pair is: both-subjects (4) with neither (8), giving 32; or Chemistry-only (6) with Physics-only (12), giving 72. Total 104. Probability $= \dfrac{104}{200} = \dfrac{13}{25}$